CIT 140 · Week 7 · Data-Quality Report

Meridian Forge · fictional inspection export v1

Learner name / date	[Enter your name and date]
Import status	20 rows loaded

Amber cells: your responses. Other results calculate from Prepared and Audit.

Rule	Check	Rows flagged	Share of all rows	**Affected Row_IDs**
Q01	Machine ID is required	2	10%	R03,R14
Q02	Work order is unique	2	10%	R05,R06
Q03	Status uses the agreed list	2	10%	R08,R16
Q04	Duration is in range	3	15%	R10,R18,R20
Q05	READY identifies an inspector	3	15%	R03,R07,R12
Distinct rows with any issue		11	55%	Count each row once, even when several Q-rules flag it. Do not add the five counts.

Scope: all 20 supplied records. Q02 flags every row in a repeated Work_Order_ID group; extra duplicate copies are a different measure. Unresolved fields remain flagged, not guessed.

Source: authored CIT140 case W07-S01 (Raw) and W07-S02 (rules). Report a data-quality finding, not a machine safety clearance. Explain your evidence and next action on page 2.

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

Row R03 fails rules Q01 and Q05 becasues its Machine_ID and Inspector_ID sre both blank, even though the status is READY and the duration is 12 minutes. This matters because an inspection record cannot be trusted or checked if we do not know which machine was tested or who did the work

Next action: what to verify, who owns it, and how to retest

Ask the Maintenance Records Owner to look at physical logs for the missing Machine_ID and ask the Shift Supervisor to confirm the Inspector_ID. Keep these fields flagged as an issue until someone verifies the missing information, then retest rues Q01 and Q05 after fixing it.

Limit: why you cannot guess a correction or prove accuracy

We cannot guess or make up a machine ID or inspector name because computer rules and text cleaning tools cannot invent real world facts. Guessing missing information creates fake evidence instead of accurately reporting real data quality.

Professional connection: one framework idea and how you used it

I used the Data Quality Framework idea of "fit for purpose". A record marked REDY is only fit for use if it clearly names the machine and the inspector responsible for work

Refresh test: before, changed value, after, and restored result

Changing Row R02 Inspection_Minutes from 20 to 0 increased the Q04 flagged count from 3 to 4. After changing it back to 20 and clicking Refresh All, the Q04 count returned to 3

Export only Report: two landscape pages, page break before row 26. Review both pages.